



MATH208

Data Structure

Szu-Chi Chung

Department of Applied Mathematics, National Sun Yat-sen University

Lectures

- Class hours: Mon. (9:10-12:00)
 - Classroom: 理 SC 2004
- Lecture: Szu-Chi Chung (鍾思齊)
 - Office: 理 SC 2002-4
 - Office hours: Mon. 16:00~18:00 and Wed. 16:00~18:00
- T.A.: 黃冠欽
 - Office: SC2003-2
 - TA hour: Wed. 16:00~17:00

Textbook and requirement

- The assignment and related material will be available on the course webpage.
Course website and Facebook group
 - <https://phonchi.github.io/nsysu-math208/>
 - <https://www.facebook.com/groups/2704670826617365>
 - <https://phonchi.github.io/ds-cpp-selfstudy/>
- Textbook: *Problem Solving with Algorithms and Data Structures using C++*
 - Authors: Brad Miller, David Ranum and Jan Pearce
- Problem Solving with Algorithms and Data Structures using Python, 3rd Edition
 - Authors: Brad Miller and David Ranum
- For the basic data structure code checkout
 - <https://github.com/phonchi/pythonds3/tree/master/cppds>

Grading policy

- Grading
 - Homework 50% (5 assignments, mainly coding parts (C++/Python))
 - Participants: 5% (participates at least 10 times can get the full score)
 - Take-home Quiz: 5% (2 times, conceptual, single choice and short answer questions)
 - Midterm exam 20% (conceptual, single choice and short answer questions)
 - Final exam 20% (conceptual, single choice and short answer questions)
- Midterm (conceptual)
 - It will be held on 2026/11/02 at 理 SC 2004
- Final (conceptual):
 - It will be held on 2026/12/21 at 理 SC 2004
- You need to use C++/Python in this course

Resources

- C++/ Python basics
 - <https://ocw.aca.ntu.edu.tw/ntu-ocw/ocw/cou/111S107> or [Python](#)
- Object-oriented
 - <https://openhome.cc/Gossip/CppGossip/> or <https://openhome.cc/zh-tw/python/>
- Practicing
 - [Hackerrank](#)
 - <https://neetcode.io/practice>
- Visualization
 - [C++/Python Tutor](#)
 - <https://visualgo.net/en>
 - <https://opensa-server.cs.vt.edu/home/about>
 - <https://cmps-people.ok.ubc.ca/ylucet/DS/Algorithms.html>

What we are going to study in this semester

- C++/ Python Object oriented fundamentals
 - Variables and containers
 - Flow Control
 - Functions
 - Encapsulation, polymorphism and inheritance (Classes)
- Algorithm analysis
 - Big O
 - Timing benchmark
- Linear structure
 - Array/Linked list
 - Stack/Queue
- Recursion
- Searching and Sorting
- Nonlinear data structure
 - Hash tables
 - Graphs
 - Trees
 - Binary search tree
 - AVL tree
- Not covered
 - Advanced graph and dynamic programming
 - Divide and conquer and Greedy algorithm
 - Network flow
 - NP and computational intractability

Relate to other courses

■ Related Courses

- Discrete Mathematics
- Algorithms
- Python and machine learning algorithms

■ Other courses

- Advance programming
- Web programming
- Network programming
- Software engineering
- Data science/Machine learning/Artificial intelligence

Some notes on using AI

- **The Shift from Determinism to Non-determinism**
 - AI models work based on probabilities, meaning the same prompt might yield different results. AI can provide code that works in one scenario but fails in another
- **The Role of AI in Development**
 - AI reduces the manual labor of typing syntax, allowing developers to focus more on high-level logic and problem-solving
 - AI should be treated as a fast but sometimes unreliable intern, so human verification is mandatory
 - Use AI for explaining complex code, generating unit tests, or suggesting alternative solutions where understanding context is key
- **Impact on Learning: Foundations Still Matter**
 - If beginners jump straight to "Vibe Coding" without understanding logic, they risk becoming "Prompt Engineers" who cannot fix systems when AI fails